

of the platform part of a PaaS style cloud can be installed on the guest operating system, PaaS clouds are easier made with virtualization!

- Since SaaS clouds have just the top layer applications added to the operating system, those too can be installed on a virtual machine OS and once again cloning can make all those come near complete automation. So SaaS style clouds can also benefit greatly from virtualization.
- Along with this, self-management can also be achieved to a large extent by just tuning the virtualization software to the specific needs or using a particularly clever one.

As you can see, all this can be done without virtualization technology, but that would be a much more complicated scene than it is with virtualization in the picture!

Types of virtualization

There are different types of virtualization. Let's briefly explore each of them.

Full virtualization

This is the form in which everything is virtualized. By everything, we mean all the hardware components visible to the guest OS, which includes the processor, storage, networking components, display etc. Most desktop solutions available for virtualization such as VirtualBox, VMware Workstation, QEMU, Parallels Workstation etc. are "Full Virtualization" solutions. They can make any guest OS run atop any host OS as long as both are supported and you don't have to worry about a thing with them.

OS level virtualization

This is a little trickier than others. In this case, a full virtual machine is not run inside a virtualization software but the application(s) is (are). For those who know, this technique is like an advanced version of the chroot command in Linux (and other UNIX-like) OS wherein the application is given an environment to work and the application is made to believe that it is (and other related components) the only thing running on the system. This is done by isolation in something called as the "user-space" environment. The best example of this is the "jailed" environment in FreeBSD. Read more about the FreeBSD jail: http://en.wikipedia.org/wiki/FreeBSD_jail

Paravirtualization

It's not possible to fully describe in the small space here what it means.

